Deliverable 2: Relational Schema, Constraints & DDL Specification

Project: Intelligence Fusion Platform • **Course:** DBMS Mini Project (UE24CS395)
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1. Schema Normalization & Design Proofs (1NF to BCNF)

1NF Proof: Every attribute contains strictly atomic, indivisible values. No multi-valued attributes or repeating groups exist.
2NF Proof: The schema is in 1NF and contains zero partial functional dependencies. In all associative tables with composite candidate keys (e.g., *case_people*, *case_vehicles*), non-prime attributes functionally depend on the entire composite key.
3NF & BCNF Proof: Every determinant is a candidate key. Transitive dependencies (e.g., storing officer department names or suspect vehicle specs redundantly inside the case table) are eliminated by isolating entities into dedicated relation schemas.

2. Integrity Constraints & Referential Actions

Constraint Class	Implementation Target	SQL Syntax Definition	Purpose
CHECK (Status)	cases.status	CHECK (status IN ('Open', 'Under Investigation', 'On Hold', 'Closed'))	Restricts status transitions
CHECK (Priority)	cases.priority	CHECK (priority IN ('Low', 'Medium', 'High', 'Critical'))	Enforces triage levels
UNIQUE Natural	officers.badge_number	UNIQUE (badge_number)	Prevents duplicate badges
UNIQUE Hardware	vehicles.vin, devices.imei	UNIQUE (vin), UNIQUE (imei)	Physical identity integrity
FOREIGN CASCADE	case_people, case_evidence	FOREIGN KEY (...) REFERENCES ... ON DELETE CASCADE	Prevents orphaned relations
FOREIGN RESTRICT	cases.lead_officer_id	FOREIGN KEY (...) REFERENCES ... ON DELETE RESTRICT	Guarantees case accountability

3. Executable DDL Excerpts (from database/sql/schema.sql)

```
-- Core Investigation Cases Table with Check Constraints & Indexes
CREATE TABLE IF NOT EXISTS cases (
    case_id VARCHAR(32) PRIMARY KEY,
    title VARCHAR(255) NOT NULL,
    description TEXT NOT NULL,
    case_type VARCHAR(100) NOT NULL,
    status VARCHAR(50) DEFAULT 'Open'
        CHECK (status IN ('Open', 'Under Investigation', 'On Hold', 'Closed')),
    priority VARCHAR(50) DEFAULT 'Medium'
        CHECK (priority IN ('Low', 'Medium', 'High', 'Critical')),
    lead_officer_id VARCHAR(32) REFERENCES officers(officer_id) ON DELETE RESTRICT,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP,
    updated_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

CREATE INDEX IF NOT EXISTS idx_cases_status ON cases(status);
CREATE INDEX IF NOT EXISTS idx_cases_priority ON cases(priority);
CREATE INDEX IF NOT EXISTS idx_cases_lead_officer ON cases(lead_officer_id);
```